

Tusoni Matteo

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ABOUT ME

My academic background originates in theoretical physics, where I developed a solid foundation in analytical reasoning and quantitative methods. I am currently pursuing a Ph.D. in Computer Science at Sapienza University of Rome, focusing on financial time series.

My research aims to understand the dynamics and interactions within complex systems, with particular attention to applications in macroeconomics and finance. I am interested in developing rigorous methods that connect formal analysis with empirical insights.

EDUCATION

- **PhD in Computer Science** *Nov 2024 - present*
Sapienza, University of Rome Rome, Italy
 - **Research Focus:** I am exploring advanced AI methodologies designed to uncover causality in time series data, with a specific focus on financial markets.
 - **Advisor:** Prof. Novella Bartolini
- **Master's degree in Theoretical Physics** *Mar 2023*
Sapienza, University of Rome Rome, Italy
 - **GPA:** 110/110 cum laude
 - **Thesis title:** Measuring the beauty and the charm of the Higgs boson: Investigating the Higgs boson decays into a pair of b- and c-quarks with the ATLAS detector at the LHC and development of machine learning methods for particle flow reconstruction.
 - **Advisor:** Prof. Marumi Kado
 - **Co-Advisor:** PhD. Francesco Armando di Bello

RESEARCH EXPERIENCE

- **Department of Computer Science, University of Oxford** *Feb 2026 - present*
Visiting PhD Student Oxford, UK
 - I am developing and analyzing differentiable approaches to agent-based model (ABM) simulations for economic and financial systems, as part of the research team led by Professor Michael Wooldridge and Professor Anisora Calinescu
- **Applied Research Team, Bank of Italy** *Jan 2025 - Aug 2025*
PhD Internship Rome, Italy
 - Engaged in the study of causal inference techniques applied to macroeconomic time series, with a focus on understanding structural dependencies in high-dimensional financial systems.
 - Involved in the exploration of multi-agent system models for macroeconomic simulations, with the aim of capturing emergent behavior and policy effects in complex adaptive economies.
 - Outcome: 2 workshop papers and a full paper under review.
- **CERN, ATLAS experiment** *Mar 2022 - Mar 2023*
Intern Geneva, Switzerland
 - Contributed to the development of an AI system for particle detection and the creation of a detector simulation. Played a role in the study of the Higgs boson within the ATLAS collaboration.
 - Active contributor for the ATLAS Note ANA-HIGG-2020-20-INT1, which summarises the measurement of Higgs bosons decaying into $b\bar{b}$ and $c\bar{c}$ in the VH production mode, representing the most recent state-of-the-art prior to the publication of final results by the entire collaboration.
 - Outcome: 2 full papers, 1 workshop paper and several Atlas collaboration internal notes.

- **LNF, Frascati National Laboratories**

Mar 2021 - Aug 2021

Intern

Frascati, Italy

- Researched plasma acceleration, investigating the interaction of an electron beam with ionized gas, as part of the SPARC_LAB collaboration.

- **INFN, National Institute of Nuclear Physics**

Mar 2021 - Mar 2023

Member

Rome, Italy

- Affiliated with the Istituto Nazionale di Fisica Nucleare (National Institute of Nuclear Physics) as a member of the Italian physics community working at CERN and LNF.

PUBLICATIONS & WORKSHOPS

- **Robust Causal Discovery in Real-World Time Series with Power-Laws**

ICML 2026 - Spotlight

2025

- Developed a robust causal discovery method leveraging power-law spectral features of time series to amplify causal signals, outperforming state-of-the-art methods on synthetic and real datasets.

- **Causal Discovery for Stochastic Power Law Time Series**

Workshop: AIFIN 2025 (ECAI 2025 Satellite Workshop)

2025

- Developed a plug-in spectral filtering framework that adapts to multiple causal discovery algorithms, leveraging power-law features to suppress noise and consistently surpass state-of-the-art results on synthetic and real datasets.

- **Placy - Power law causal Discovery**

Workshop: Complexity in Economics & Finance (StatPhys29 Satellite Workshop)

2025

- Presented PLACy, our newly developed causal discovery algorithm.

- **Reconstructing particles in jets using set transformer and hypergraph prediction networks**

Published in: European Physical Journal, Q1 classification.

2023

- Development of three separate set-to-set neural network architectures to reconstruct particles in events containing a single jet in a fully-simulated calorimeter.

- **Configurable calorimeter simulation for AI applications**

Published in: Machine Learning: Science and Technology, Q1 classification

2023

- A configurable calorimeter simulation aimed to support the development of machine learning algorithms in high energy physics that rely on realistic particle shower descriptions, such as reconstruction, fast simulation, and low-level analysis.

- **HGPflow: Particle reconstruction as hyperedge prediction**

Workshop: Machine Learning and the Physical Sciences (NeurIPS 2022 Satellite Workshop)

2022

- Modeling of particle reconstruction as a set-to-set mapping over hypergraph incidence structures, facilitating accurate and interpretable momentum estimation with built-in energy conservation.

AWARDS

- **G-Research — Spring into Quant Finance**

Winner

Apr 2026

- Awarded a place on G-Research's week-long residential programme introducing quantitative finance.

- **Sapienza — Avvio alla ricerca**

Winner

Jan 2026

- Sapienza University of Rome's internal seed-funding scheme for early-career researchers. It supports young researchers in starting independent research projects.

PROFESSIONAL EXPERIENCE

- **Ernest & Young S.p.a**
Senior Data Scientist
 - **Leonardo S.p.a**
Data Scientist and AI Specialist

May 2024 - Nov 2024
Rome, Italy

Jun 2023 - May 2024
Rome, Italy

SKILLS

- **Programming Languages:** C, C++, Python, R, Julia
- **Operating Systems:** Windows, MacOS, and Linux
- **Tools and Platforms:** MATLAB, SQL, Docker, Git, Gnuplot, Root, TensorFlow, PyTorch, scikit-learn, Keras
- **Languages:** Italian (Native), English (Advanced), French (Basic understanding)